

SOTA FUJII

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EDUCATION

DePauw University

August 2023 - Present (Expected Graduation: May 2027)

B.A. in Computer Science and Business Analytics | GPA: 3.73/4.00

Greencastle, Indiana, United States

- Relevant Courses: Advanced Calculus, Linear Algebra, Advanced Probability and Statistics, Data Structures and Algorithms
- Awards & Honors: Presidential Award of Excellence (Merit scholarship), Dean's List (All eligible semesters)

University of Oxford, Jesus College

January 2026 - June 2026

Visiting Student | First-Class Performance

Oxford, Oxfordshire, United Kingdom

- Relevant Tutorials: Artificial Intelligence, Statistical Machine Learning, Deep Learning for NLP, Information Theory

RESEARCH EXPERIENCE

Identifiability of Score-Based Generative Models (DePauw University)

August 2026 - Present

Undergraduate Researcher | Supervisor: Prof. Brian Howard

Greencastle, Indiana, United States

- Developed an identifiability theory for diffusion models, proving that forward noising cannot create latent identifiability and that identifiability from conditioning is preserved
- Derived an explicit, parameter-free readout of latent sufficient statistics from conditional score contrasts, with quantitative recovery guarantees linking representation error to the score-matching objective
- Extended the theory to discrete diffusion and auxiliary-variable-free settings, establishing exact recovery below an explicit error threshold in the discrete case

Domain-Specific Embedding Model for RAG (Menicon Co., Ltd.)

June 2026 - August 2026

ML Research Engineer Intern

Nagoya, Aichi, Japan (Hybrid)

- Fine-tuned an open-weight embedding model using Low-Rank Adaptation (LoRA), domain-adapted BERT distillation, proximal policy optimization (PPO) reinforcement learning, and proprietary corpora for retrieval in a specialized RAG pipeline
- Conducted 300+ hyperparameter experiments across model and training configurations, improving retrieval and sentence-similarity benchmark performance by more than 20% over the SOTA embedding model

Interpretable Machine Learning for Birth-Outcome Analysis (DePauw University)

May 2025 - September 2025

Student Research Assistant | Supervisor: Prof. Yanchao Yang

Greencastle, Indiana, United States

- Analyzed 2011–2021 birth records in the United States to study C-section delivery, low birthweight, and prematurity across Hispanic-origin groups using logistic regression and classification trees
- Conducted literature review, large-scale data preprocessing (over 30 million individual records), statistical modeling, and interpretation, contributing to a peer-reviewed publication

PUBLICATIONS & MANUSCRIPTS

- [Fujii, S.](#) (2026). "Identifiability of the Diffusion Models: The Conservation Law and A Recovery Map." *AISTATS 2027 Main Conference*. (Under review)
- [Fujii, S.](#) (2026). "A Conservation Law for Identifiability in Diffusion Models: Noising Creates Nothing, Destroys Nothing." *AI for Stochastic Dynamics Workshop at NeurIPS 2026*. (Under review)
- Yang, Y., [Fujii, S.](#), & Nguyen, T. (2025). "Birth Outcomes in the Hispanic Population in the United States: Trends, Variation, and Determinants (2011–2021)." *International Journal of Environmental Research and Public Health*, 22(9), 1325. (Peer-reviewed)

INDUSTRY EXPERIENCE

Rubyin Corp.

August 2024 - Present

Software Engineer Intern

Japan (Remote)

- Built an AI-powered application using the Claude API and implemented a Model Context Protocol (MCP) server to connect the model with external tools and application data

Sekisho Corp.

July 2024 - July 2024

Analytics Engineer Intern

Tsukuba, Ibaraki, Japan

- Developed a Bayesian statistical classifier prototype to flag risks from customer inquiries using a Japanese NLP framework

SKILLS

Programming Languages: Python, TypeScript, SQL, C++

Frameworks & Tools: PyTorch, Sentence-Transformers, MLflow, GCP, Git, FastAPI, Cloudflare, Supabase, Stata

Languages: English (Proficient), Japanese (Native)